

Assignment 4: Assignment 4: Securing a Cloud API

Course: Cloud Computing & Data Engineering

By: Hamed Ahmadinia

Introduction

In Assignment 4, I secured a cloud-deployed REST API using FastAPI by applying secrets management and authorization techniques. The work builds on the Docker-GitHub-cloud deployment pipeline from Assignment 2 and focuses on protecting API endpoints in both local and deployed environments.

The assignment consists of three parts. In Part A, I protected an endpoint using an API key. In Part B, I deployed the application to Render and managed secrets using environment variables. In Part C, I added JWT-based authentication with role-based authorization to control access to user-specific data.

Overall, the assignment demonstrates how different security mechanisms, such as API keys and JWTs, can be used to protect APIs depending on access requirements. The final application includes a public endpoint, an API key-protected endpoint, and a JWT-protected endpoint, all tested locally and in the cloud.

Part A – FastAPI API Key Protected Endpoint

In Part A of the assignment, an existing FastAPI project repository was extended to include an API key-protected endpoint. The project was based on the docker-fastapi template and was hosted in a public GitHub repository named **paas-fastapi-Cloud-Computing**. The repository already contained the required FastAPI application structure, including the app/ directory, Dockerfile, docker-compose.yml, requirements.txt, and supporting configuration files. This repository served as the foundation for implementing API authentication and access control.

The main development work was performed in the app/main.py file. A new dependency function, require_api_key, was implemented using FastAPI's dependency injection system. This function reads an expected API key from an environment variable (API_KEY) and validates incoming requests by checking the X-API-Key request header. If the API key is missing or invalid, the application returns a 401 Unauthorized response. A public root endpoint (/) was also maintained to allow unauthenticated access and to verify that the application was running correctly.

A protected endpoint (/secret) was added and configured to require a valid API key. This endpoint was tested locally using the FastAPI Swagger UI and direct browser access. When accessed without the required header, the endpoint correctly returned a 401 Unauthorized error. When the correct API key was provided via the X-API-Key header in Swagger UI, the endpoint returned the expected protected response, confirming that the authentication mechanism was functioning as intended.

Issues Encountered

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Several issues were encountered during local development and testing:

- The uvicorn command was initially not recognized due to environment path conflicts between the Microsoft Store Python installation and the Anaconda Python environment.
- The FastAPI server did not always restart automatically when code changes were made, leading to confusion about whether updates were being applied.
- Requests to the /secret endpoint returned 401 Unauthorized responses until it was confirmed that the API key must be provided using the X-API-Key header rather than as a query parameter.
- Docker-related commands (docker ps, docker compose down) failed locally because Docker Desktop was not running, although Docker was not required for completing Part A.

Resolution

These issues were resolved by explicitly running Uvicorn using the Anaconda Python interpreter, ensuring the correct environment was used. The --reload flag was used to enable automatic server reloads during development. The API key was tested correctly via Swagger UI by supplying it in the request headers, which confirmed the expected behavior of the protected endpoint. Docker errors were identified as unrelated to Part A and were safely ignored at this stage.

After successful testing, the changes were committed and pushed to GitHub with the commit message **"Assignment 4: API key protected endpoint"**. The repository was verified to ensure that the .env file containing the secret API key was not committed, while an .env-example file remained available for documentation purposes. This completed Part A and prepared the project for deployment in subsequent parts of the assignment.

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The screenshot shows the GitHub repository page for 'paas-fastapi-Cloud-Computing' by Hamed-Ahmadinia. The repository is public and has 2 commits. The file list includes: app, env-example, gitignore, Dockerfile, README.md, docker-compose.yml, and requirements.txt. The README file is selected, showing the title 'docker-fastapi' and a description: 'A sample project for deploying to a PaaS Service like Render or CSC Rahti.' It also includes instructions for deployment to Render, such as logging in to <https://render.com/> and creating a new web service. The right sidebar shows the repository's activity, including stars, forks, and releases.

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

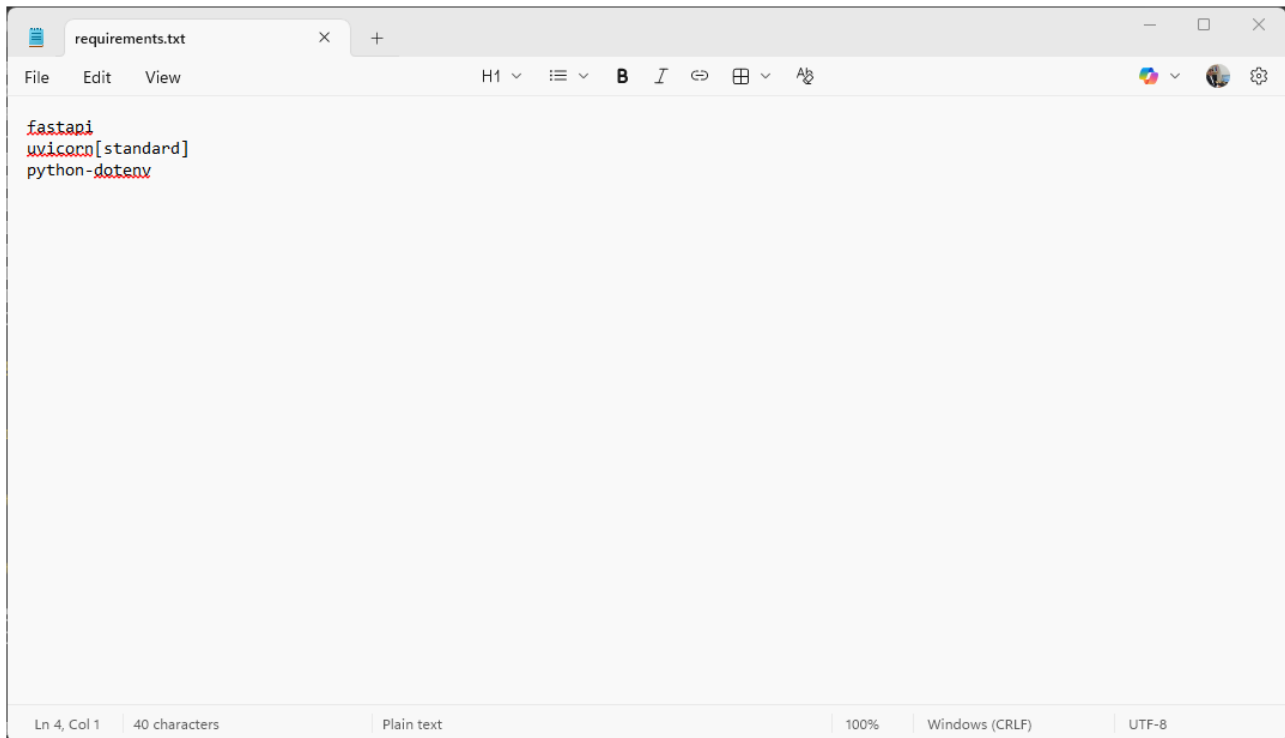
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\hamed> git clone https://github.com/Hamed-Ahmadinia/paas-fastapi-Cloud-Computing.git
Cloning into 'paas-fastapi-Cloud-Computing'...
remote: Enumerating objects: 15, done.
remote: Counting objects: 100% (15/15), done.
remote: Compressing objects: 100% (10/10), done.
remote: Total 15 (delta 2), reused 13 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (15/15), 4.19 KiB | 2.09 MiB/s, done.
Resolving deltas: 100% (2/2), done.
PS C:\Users\hamed> cd paas-fastapi-Cloud-Computing
PS C:\Users\hamed\paas-fastapi-Cloud-Computing>
```

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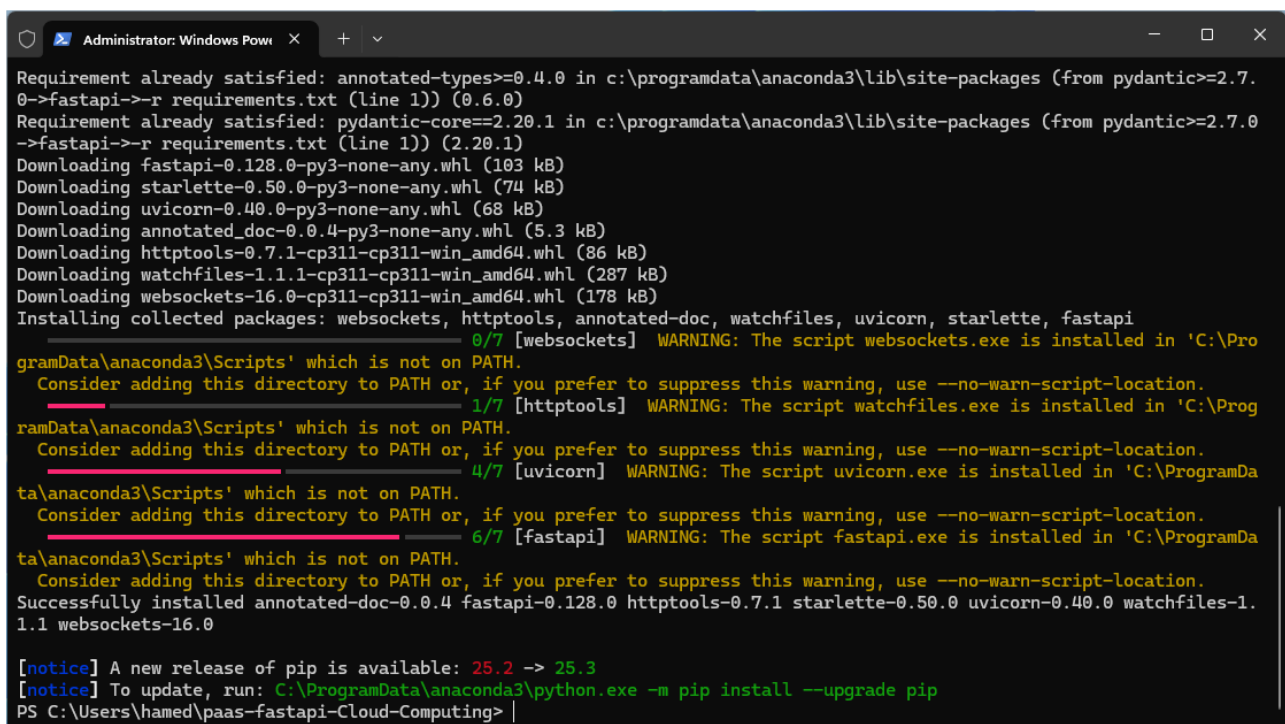
By: Hamed Ahmadiania



A screenshot of a text editor window titled 'requirements.txt'. The editor shows the following content:

```
fastapi
uvicorn[standard]
python-dotenv
```

The status bar at the bottom indicates 'Ln 4, Col 1', '40 characters', 'Plain text', '100%', 'Windows (CRLF)', and 'UTF-8'.



A screenshot of a Windows PowerShell terminal window titled 'Administrator: Windows PowerShell'. The terminal shows the execution of the command `pip install -r requirements.txt` and the output:

```
Requirement already satisfied: annotated-types>=0.4.0 in c:\programdata\anaconda3\lib\site-packages (from pydantic>=2.7.0->fastapi->-r requirements.txt (line 1)) (0.6.0)
Requirement already satisfied: pydantic-core==2.20.1 in c:\programdata\anaconda3\lib\site-packages (from pydantic>=2.7.0->fastapi->-r requirements.txt (line 1)) (2.20.1)
Downloading fastapi-0.128.0-py3-none-any.whl (103 kB)
Downloading starlette-0.50.0-py3-none-any.whl (74 kB)
Downloading uvicorn-0.40.0-py3-none-any.whl (68 kB)
Downloading annotated_doc-0.0.4-py3-none-any.whl (5.3 kB)
Downloading httptools-0.7.1-cp311-cp311-win_amd64.whl (86 kB)
Downloading watchfiles-1.1.1-cp311-cp311-win_amd64.whl (287 kB)
Downloading websockets-16.0-cp311-cp311-win_amd64.whl (178 kB)
Installing collected packages: websockets, httptools, annotated-doc, watchfiles, uvicorn, starlette, fastapi
0/7 [websockets] WARNING: The script websockets.exe is installed in 'C:\ProgramData\anaconda3\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
1/7 [httptools] WARNING: The script watchfiles.exe is installed in 'C:\ProgramData\anaconda3\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
4/7 [uvicorn] WARNING: The script uvicorn.exe is installed in 'C:\ProgramData\anaconda3\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
6/7 [fastapi] WARNING: The script fastapi.exe is installed in 'C:\ProgramData\anaconda3\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed annotated-doc-0.0.4 fastapi-0.128.0 httptools-0.7.1 starlette-0.50.0 uvicorn-0.40.0 watchfiles-1.1.1 websockets-16.0

[notice] A new release of pip is available: 25.2 -> 25.3
[notice] To update, run: C:\ProgramData\anaconda3\python.exe -m pip install --upgrade pip
PS C:\Users\hamed\paas-fastapi-Cloud-Computing>
```

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```
Administrator: Windows Powe x + v
0/7 [websockets] WARNING: The script websockets.exe is installed in 'C:\Pro
gramData\anaconda3\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
1/7 [httptools] WARNING: The script watchfiles.exe is installed in 'C:\Prog
ramData\anaconda3\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
4/7 [uvicorn] WARNING: The script uvicorn.exe is installed in 'C:\ProgramDa
ta\anaconda3\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
6/7 [fastapi] WARNING: The script fastapi.exe is installed in 'C:\ProgramDa
ta\anaconda3\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed annotated-doc-0.0.4 fastapi-0.128.0 httptools-0.7.1 starlette-0.50.0 uvicorn-0.40.0 watchfiles-1.
1.1 websockets-16.0

[notice] A new release of pip is available: 25.2 -> 25.3
[notice] To update, run: C:\ProgramData\anaconda3\python.exe -m pip install --upgrade pip
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> uvicorn app.main:app --reload
uvicorn : The term 'uvicorn' is not recognized as the name of a cmdlet, function, script file, or operable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ uvicorn app.main:app --reload
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (uvicorn:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\hamed\paas-fastapi-Cloud-Computing> python -m uvicorn app.main:app --reload
C:\Users\hamed\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.12_qbz5n2kfra8p0\python.exe: No mod
ule named uvicorn
PS C:\Users\hamed\paas-fastapi-Cloud-Computing>
```

```
Administrator: Windows Powe x + v
ta\anaconda3\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
6/7 [fastapi] WARNING: The script fastapi.exe is installed in 'C:\ProgramDa
ta\anaconda3\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed annotated-doc-0.0.4 fastapi-0.128.0 httptools-0.7.1 starlette-0.50.0 uvicorn-0.40.0 watchfiles-1.
1.1 websockets-16.0

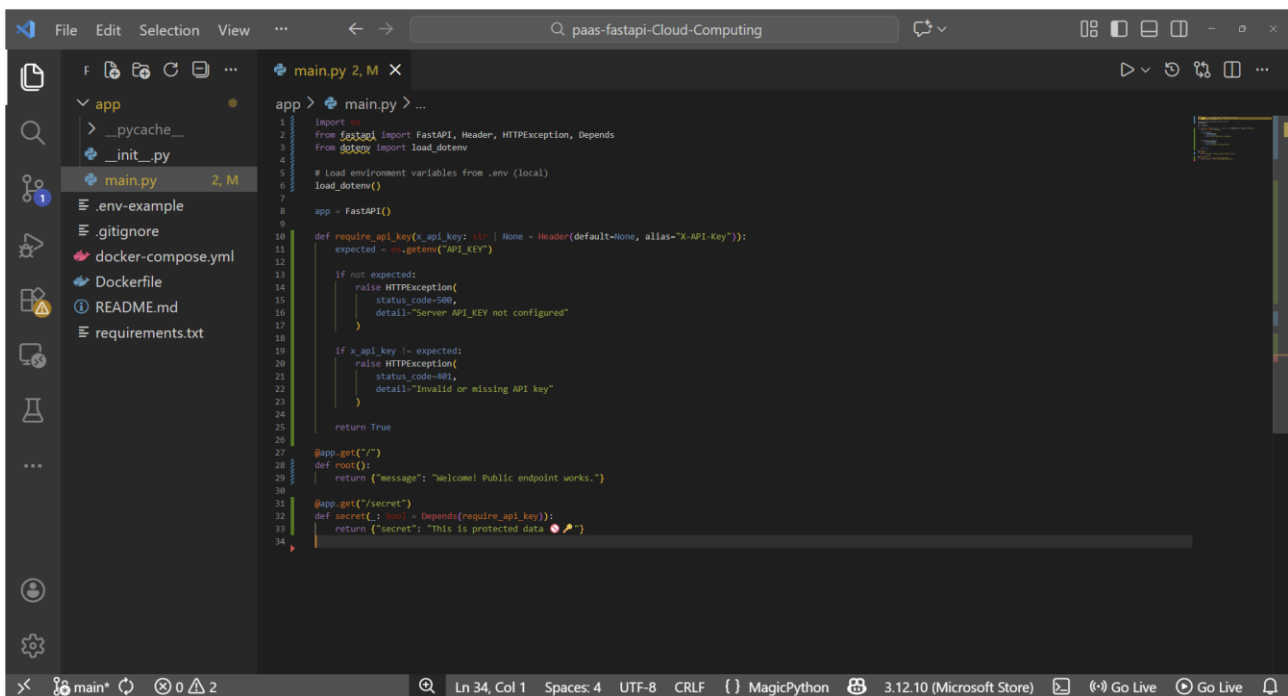
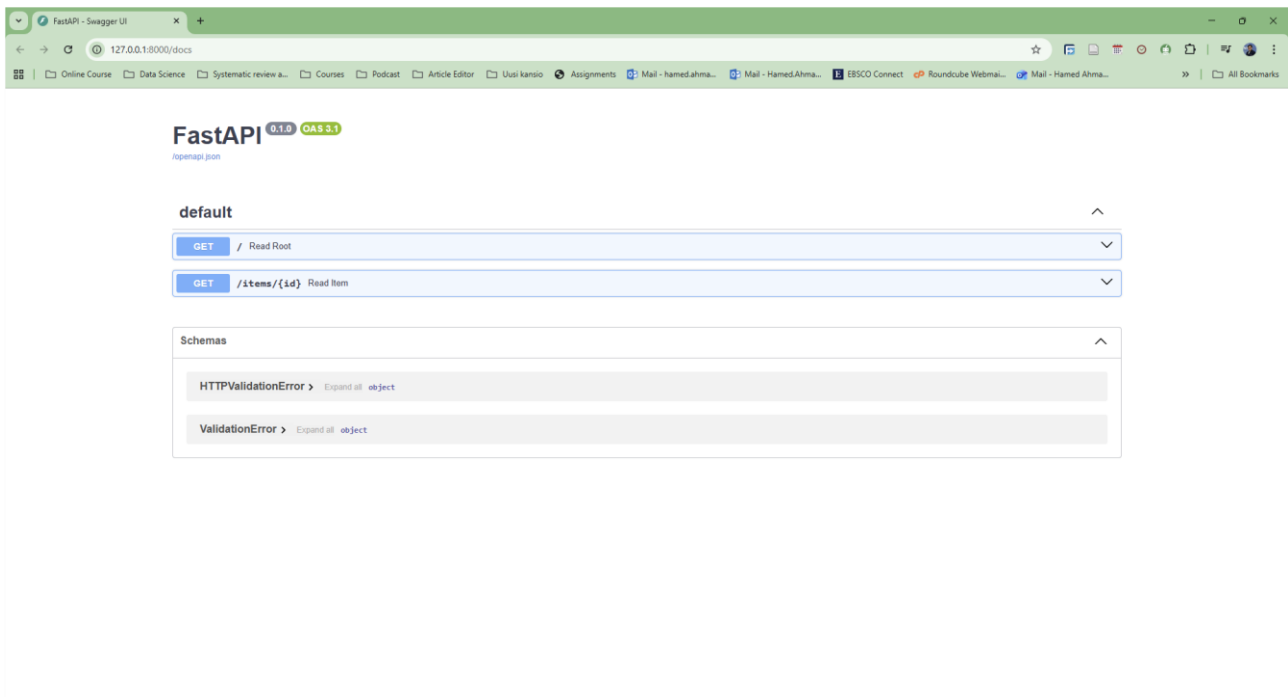
[notice] A new release of pip is available: 25.2 -> 25.3
[notice] To update, run: C:\ProgramData\anaconda3\python.exe -m pip install --upgrade pip
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> uvicorn app.main:app --reload
uvicorn : The term 'uvicorn' is not recognized as the name of a cmdlet, function, script file, or operable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ uvicorn app.main:app --reload
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (uvicorn:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\hamed\paas-fastapi-Cloud-Computing> python -m uvicorn app.main:app --reload
C:\Users\hamed\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.12_qbz5n2kfra8p0\python.exe: No mod
ule named uvicorn
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> C:\ProgramData\anaconda3\python.exe -m uvicorn app.main:app --reload
INFO: Will watch for changes in these directories: ['C:\\Users\\hamed\\paas-fastapi-Cloud-Computing']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [25064] using WatchFiles
INFO: Started server process [24384]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

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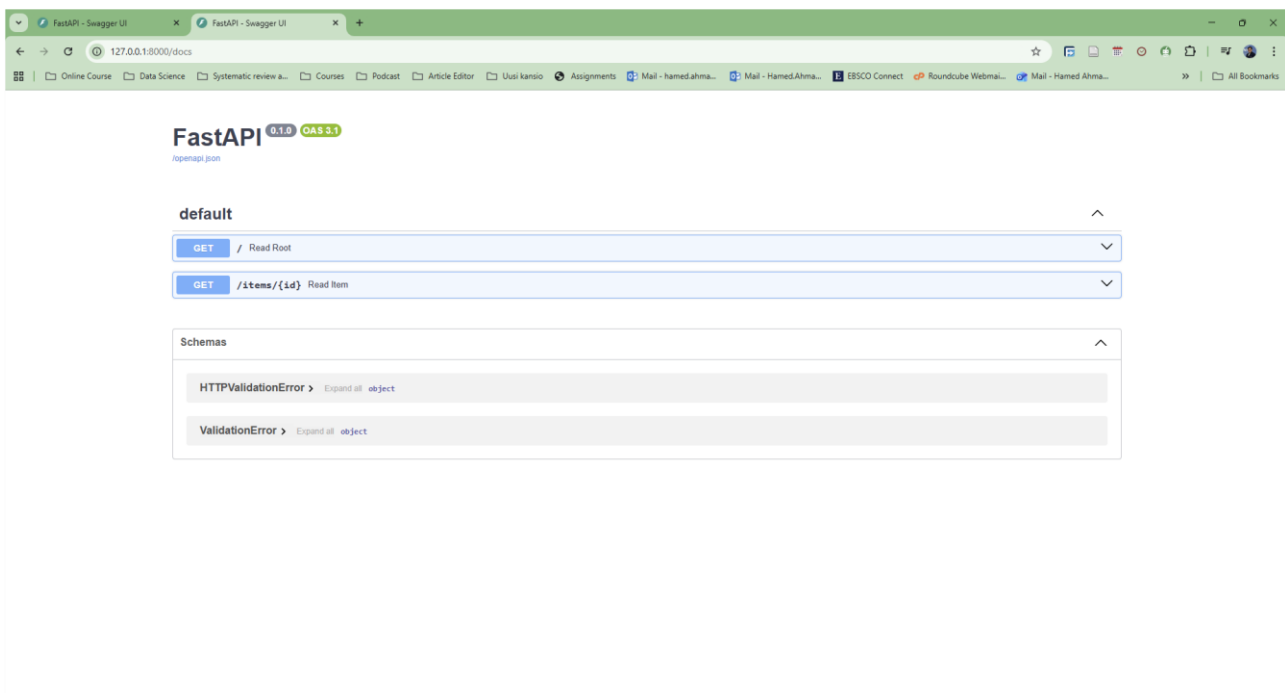


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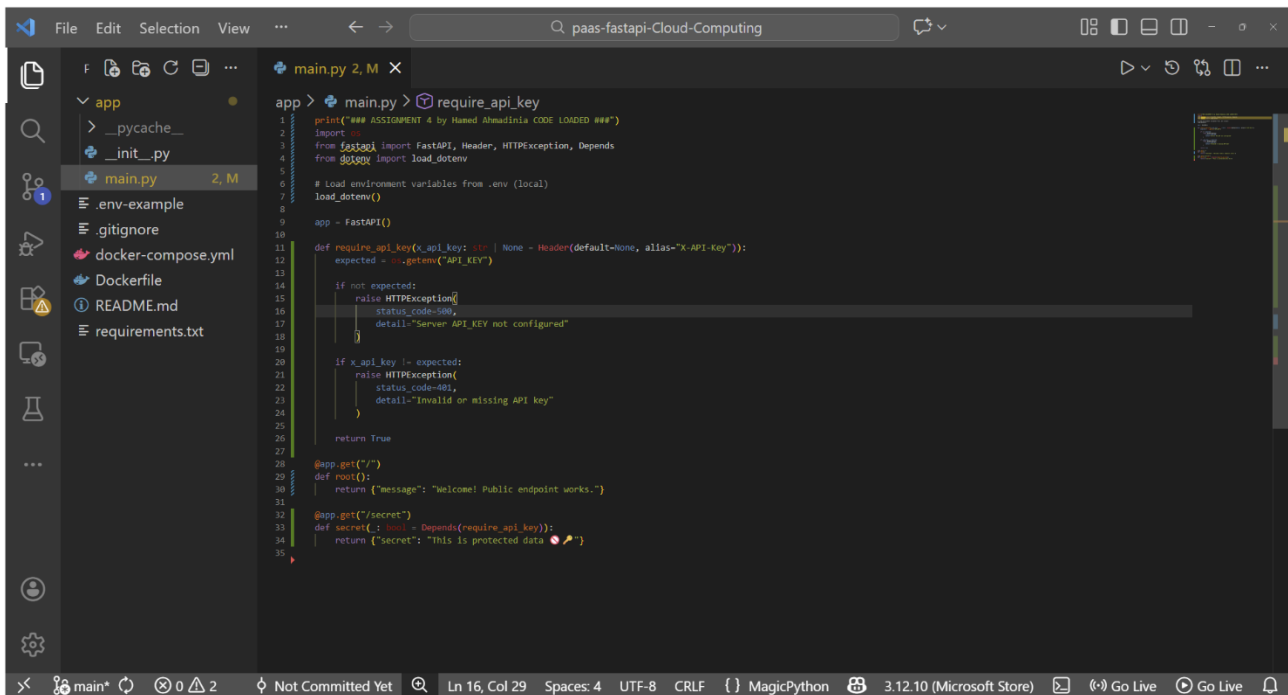
```
Administrator: Windows PowerShell
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> python -m uvicorn app.main:app --reload
C:\Users\hamed\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.12_qbz5n2kfra8p0\python.exe: No module named uvicorn
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> C:\ProgramData\anaconda3\python.exe -m uvicorn app.main:app --reload
INFO: Will watch for changes in these directories: ['C:\\Users\\hamed\\paas-fastapi-Cloud-Computing']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [25064] using WatchFiles
INFO: Started server process [24384]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:62475 - "GET /docs HTTP/1.1" 200 OK
INFO: 127.0.0.1:62475 - "GET /openapi.json HTTP/1.1" 200 OK
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [24384]
INFO: Stopping reloader process [25064]
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> ^C
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> ^C
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> ^C
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> ^C
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> C:\ProgramData\anaconda3\python.exe -m uvicorn app.main:app --reload
INFO: Will watch for changes in these directories: ['C:\\Users\\hamed\\paas-fastapi-Cloud-Computing']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [7996] using WatchFiles
INFO: Started server process [18540]
INFO: Waiting for application startup.
INFO: Application startup complete.
```



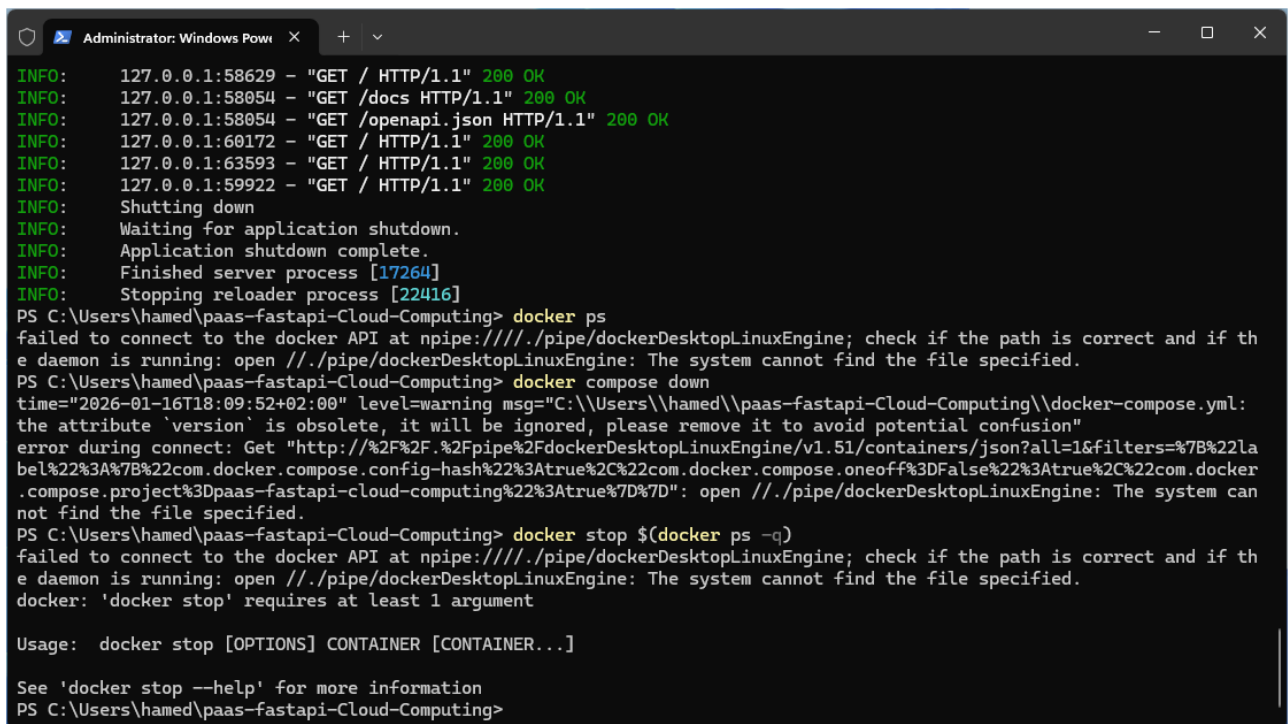
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```
File Edit Selection View ... paas-fastapi-Cloud-Computing
main.py 2, M X
app > main.py > require_api_key
1 print("### ASSIGNMENT 4 by Hamed Ahmadinia (CODE LOADED) ###")
2 import os
3 from fastapi import FastAPI, Header, HTTPException, Depends
4 from dotenv import load_dotenv
5
6 # Load environment variables from .env (local)
7 load_dotenv()
8
9 app = FastAPI()
10
11 def require_api_key(x_api_key: str | None = Header(default=None, alias="X-API-Key")):
12     expected = os.getenv("API_KEY")
13
14     if not expected:
15         raise HTTPException(
16             status_code=500,
17             detail="Server API_KEY not configured"
18         )
19
20     if x_api_key != expected:
21         raise HTTPException(
22             status_code=401,
23             detail="Invalid or missing API key"
24         )
25
26     return True
27
28 @app.get("/")
29 def root():
30     return {"message": "Welcome! Public endpoint works."}
31
32 @app.get("/secret")
33 def secret(x_api_key: str = Depends(require_api_key)):
34     return {"secret": "This is protected data 🚀"}
35
```

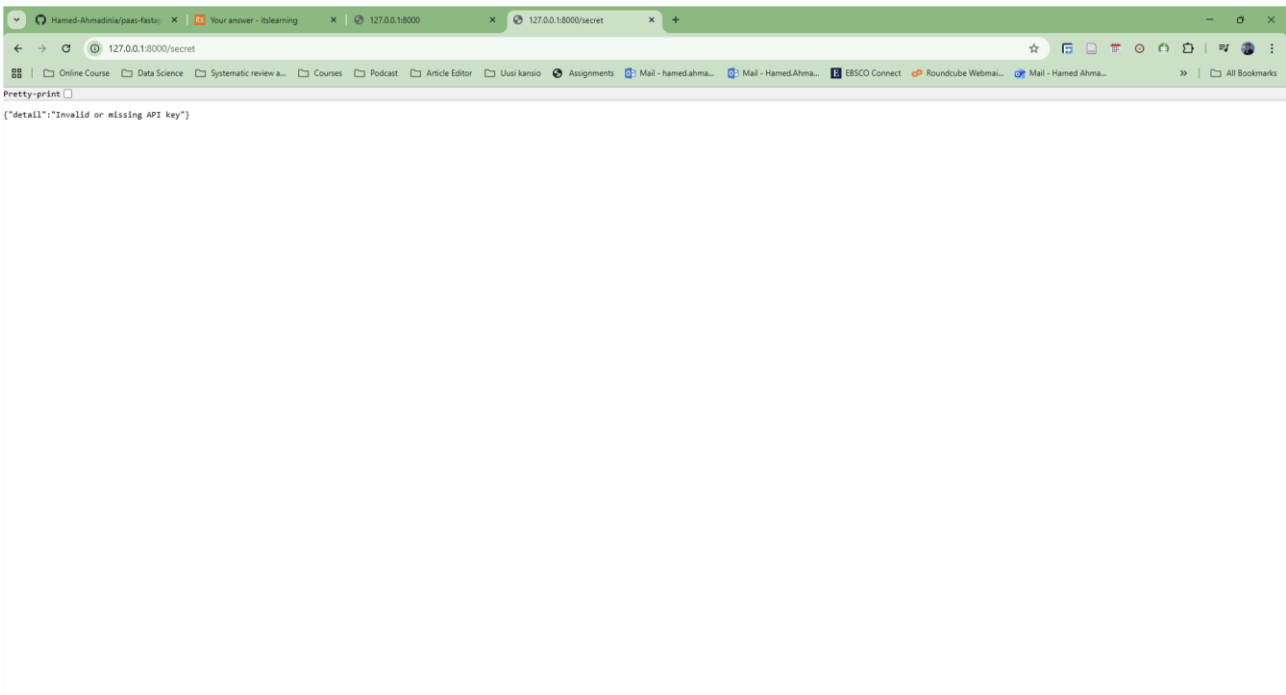
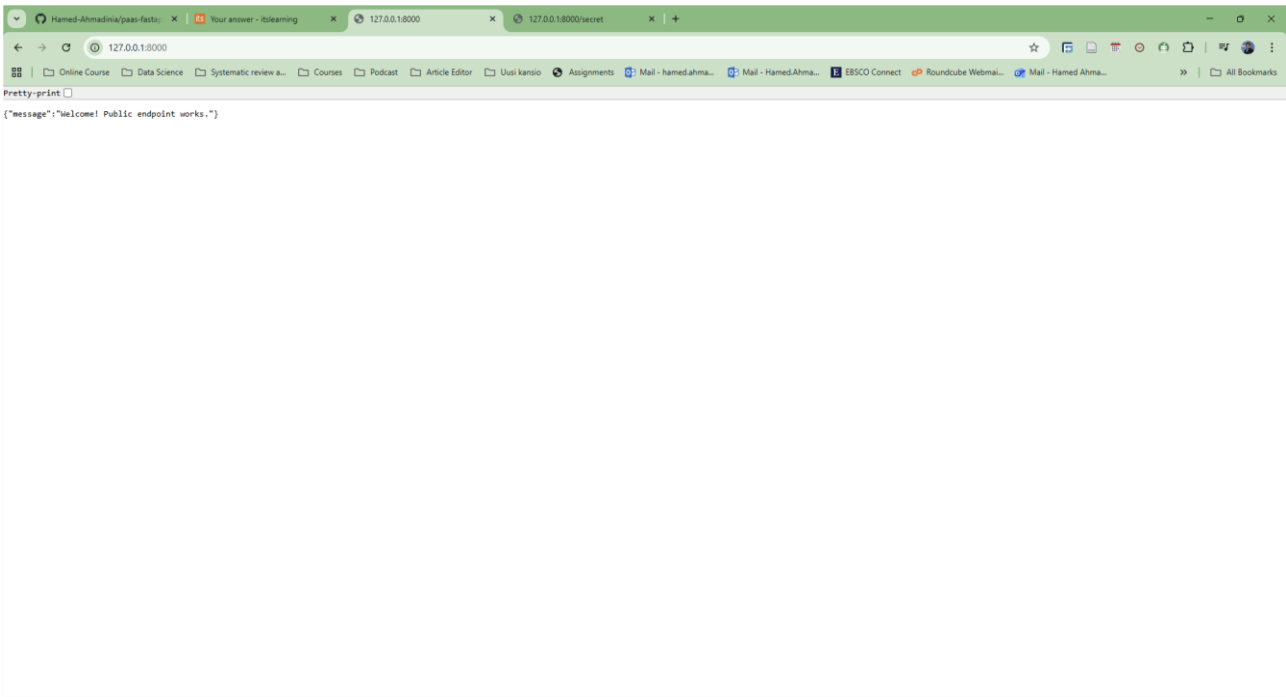


```
Administrator: Windows PowerShell
INFO: 127.0.0.1:58629 - "GET / HTTP/1.1" 200 OK
INFO: 127.0.0.1:58054 - "GET /docs HTTP/1.1" 200 OK
INFO: 127.0.0.1:58054 - "GET /openapi.json HTTP/1.1" 200 OK
INFO: 127.0.0.1:60172 - "GET / HTTP/1.1" 200 OK
INFO: 127.0.0.1:63593 - "GET / HTTP/1.1" 200 OK
INFO: 127.0.0.1:59922 - "GET / HTTP/1.1" 200 OK
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [17264]
INFO: Stopping reloader process [22416]
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> docker ps
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> docker compose down
time="2026-01-16T18:09:52+02:00" level=warning msg="C:\\Users\\hamed\\paas-fastapi-Cloud-Computing\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
error during connect: Get "http://%2F%2F.%2Fpipe%2FdockerDesktopLinuxEngine/v1.51/containers/json?all=1&filters=%7B%22label%22%3A%7B%22com.docker.compose.config-hash%22%3Atrue%2C%22com.docker.compose.oneoff%3Dfalse%22%3Atrue%2C%22com.docker.compose.project%3Dpaas-fastapi-cloud-computing%22%3Atrue%7D%7D": open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> docker stop $(docker ps -q)
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if the daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.
docker: 'docker stop' requires at least 1 argument

Usage: docker stop [OPTIONS] CONTAINER [CONTAINER...]

See 'docker stop --help' for more information
PS C:\Users\hamed\paas-fastapi-Cloud-Computing>
```

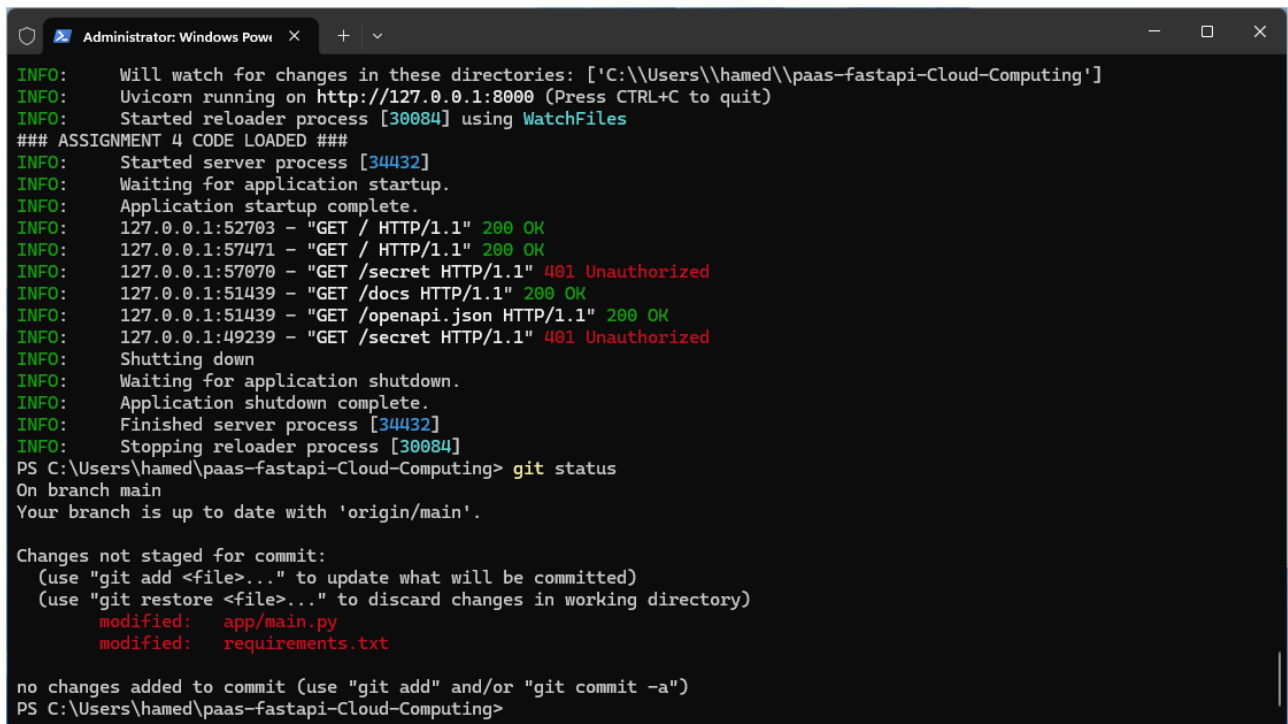
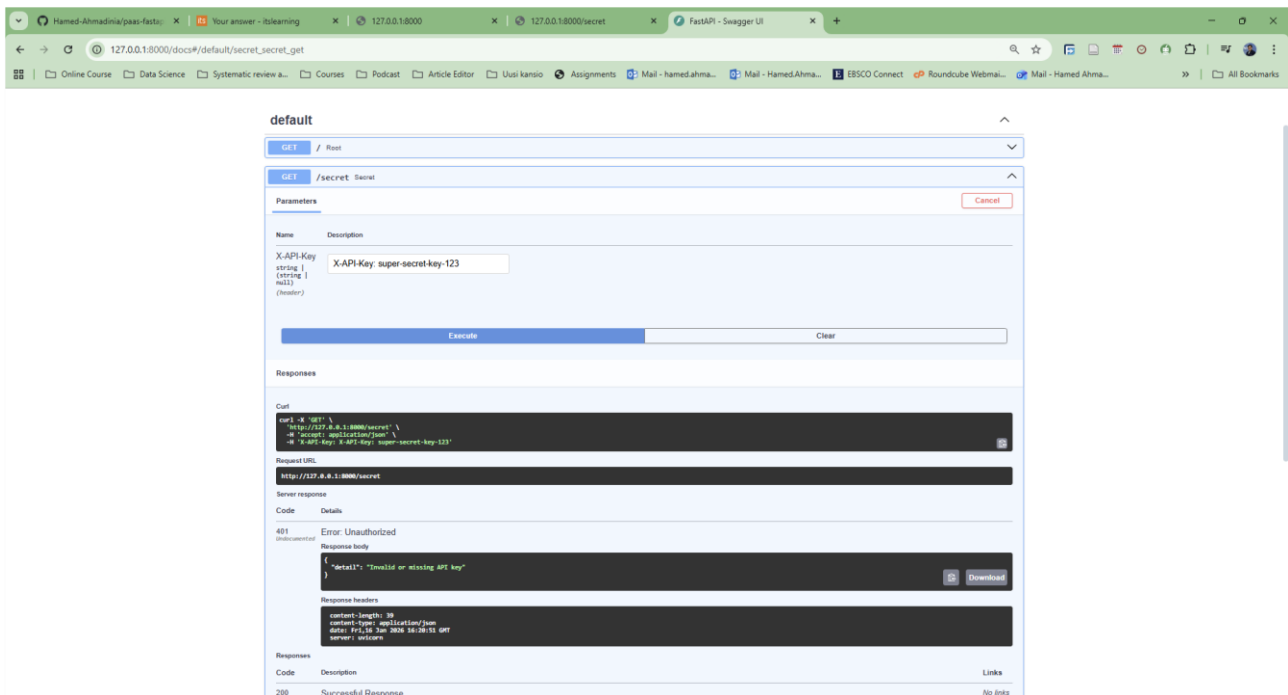

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```
Administrator: Windows PowerShell
INFO: 127.0.0.1:51439 - "GET /docs HTTP/1.1" 200 OK
INFO: 127.0.0.1:51439 - "GET /openapi.json HTTP/1.1" 200 OK
INFO: 127.0.0.1:49239 - "GET /secret HTTP/1.1" 401 Unauthorized
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [34432]
INFO: Stopping reloader process [30084]
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   app/main.py
        modified:   requirements.txt

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> git add app/main.py requirements.txt
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   app/main.py
        modified:   requirements.txt

PS C:\Users\hamed\paas-fastapi-Cloud-Computing>
```

```
Administrator: Windows PowerShell
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   app/main.py
        modified:   requirements.txt

no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> git add app/main.py requirements.txt
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> git status
On branch main
Your branch is up to date with 'origin/main'.

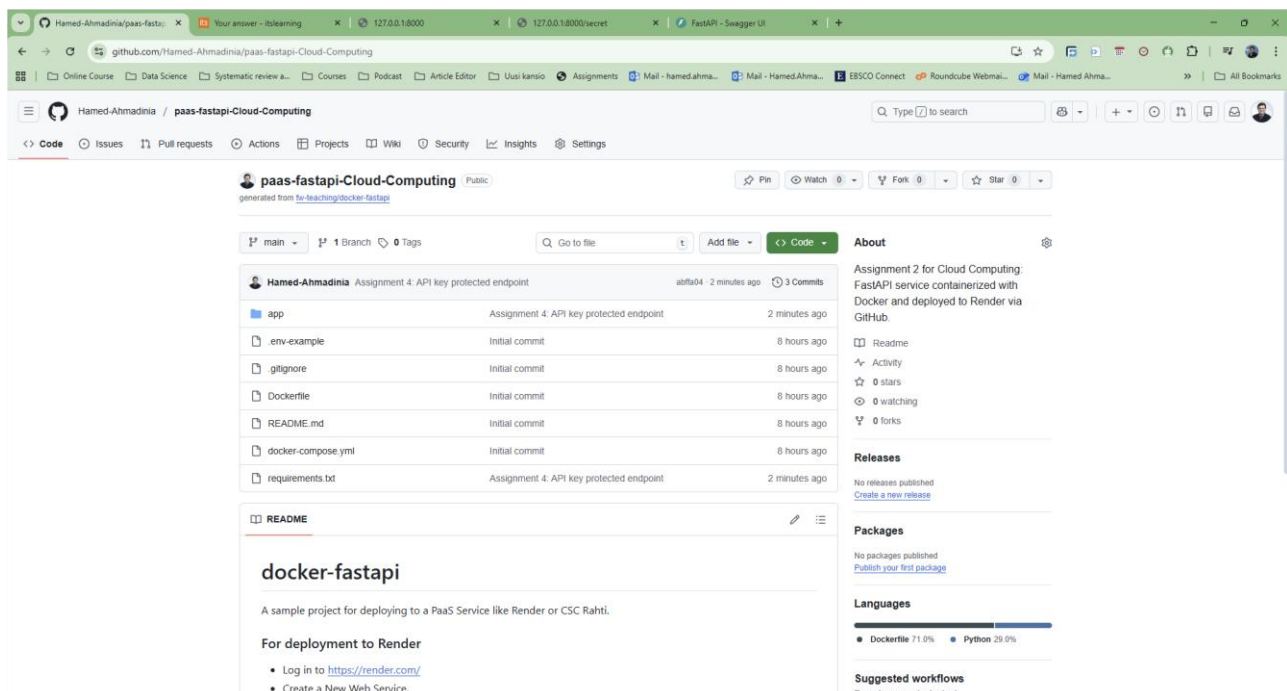
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   app/main.py
        modified:   requirements.txt

PS C:\Users\hamed\paas-fastapi-Cloud-Computing> git commit -m "Assignment 4: API key protected endpoint"
[main abffa04] Assignment 4: API key protected endpoint
 2 files changed, 25 insertions(+), 9 deletions(-)
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> git push
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 20 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 885 bytes | 885.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Hamed-Ahmadinia/paas-fastapi-Cloud-Computing.git
 860e95e..abffa04  main -> main
PS C:\Users\hamed\paas-fastapi-Cloud-Computing>
```

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Part B – Deployment and Environment Variables (Render)

In Part B of the assignment, the FastAPI application developed in Part A was deployed to a cloud service using Render. The existing GitHub repository (paas-fastapi-Cloud-Computing) was connected to Render, and the service was configured to build and run the application using a Docker-based workflow, consistent with the deployment pipeline used in Assignment 2. Render automatically cloned the repository and built the application using the provided Dockerfile.

During the initial deployment attempts, the service failed to start correctly. Investigation of the Render logs revealed that the application was using the FastAPI CLI (fastapi run / fastapi dev) and a hardcoded port, which is not compatible with Render's runtime environment. Additionally, the required FastAPI standard dependencies were missing, causing runtime errors during startup.

To resolve these issues, the deployment configuration was adjusted as follows. The requirements.txt file was updated to include fastapi[standard], ensuring that all required FastAPI dependencies were available in the production environment. The Dockerfile was also

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modified to use a production-safe startup command based on Uvicorn rather than the FastAPI CLI. The application was configured to bind to 0.0.0.0 and listen on the port provided by Render via the \$PORT environment variable. After committing and pushing these changes to GitHub, the service was redeployed successfully and reached a “Deploy live” state.

Once the application was running in the cloud, secrets management was configured using Render’s environment variable system. An environment variable named API_KEY was added through the Render dashboard, storing a randomly generated API key value. The service was redeployed to ensure the environment variable was available at runtime. This approach ensured that sensitive information was not stored in the source code or committed to the GitHub repository.

The deployed API was then tested using the public Render URL. Accessing the public root endpoint (/) returned the expected welcome message, confirming that the application was running correctly. Accessing the protected /secret endpoint without providing an API key returned a 401 Unauthorized response, as expected. When the correct API key was supplied via the X-API-Key request header using the FastAPI Swagger UI, the endpoint returned a 200 OK response with the protected data. These tests confirmed that the API key authorization mechanism worked correctly in the deployed environment.

Issues Encountered

Several issues were encountered during deployment:

- The application initially failed to start on Render due to the use of the FastAPI CLI and a hardcoded port, which are not supported in Render’s production environment.
- The deployment logs indicated missing FastAPI standard dependencies, resulting in runtime errors.
- After adding the API key in the Render dashboard, the application initially returned a “Server API_KEY not configured” error because the service had not yet been redeployed with the new environment variable.

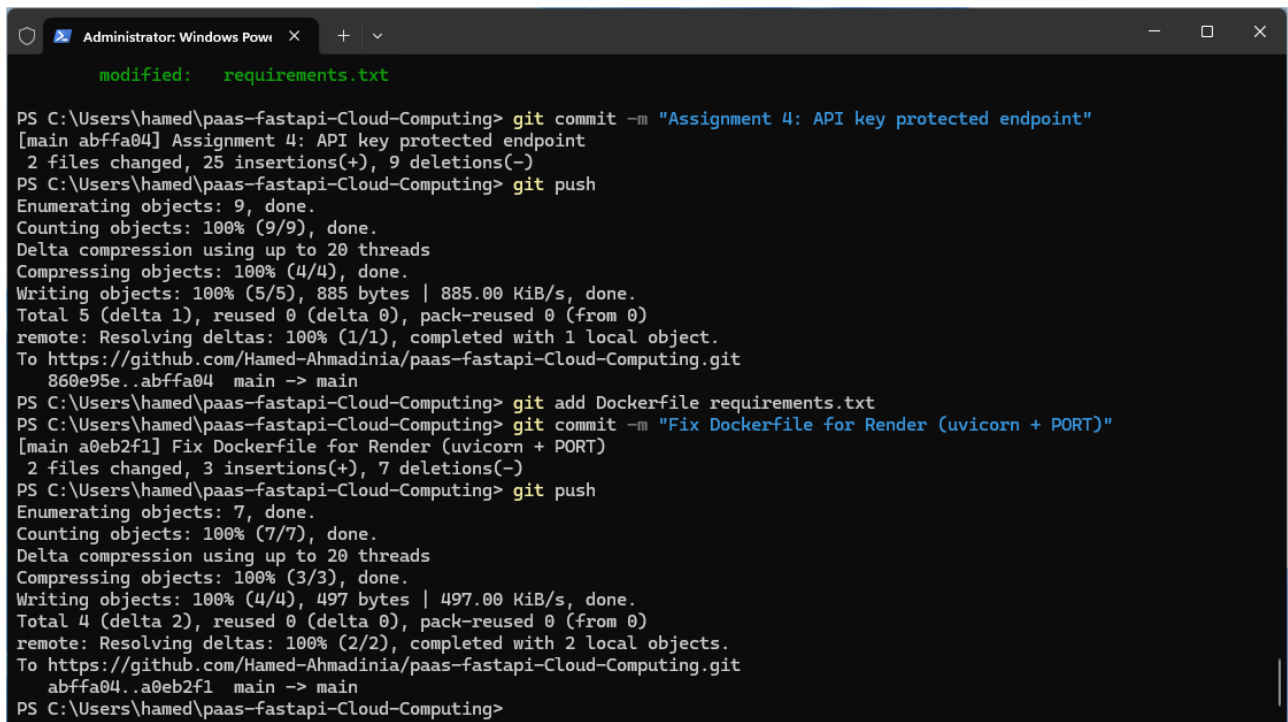
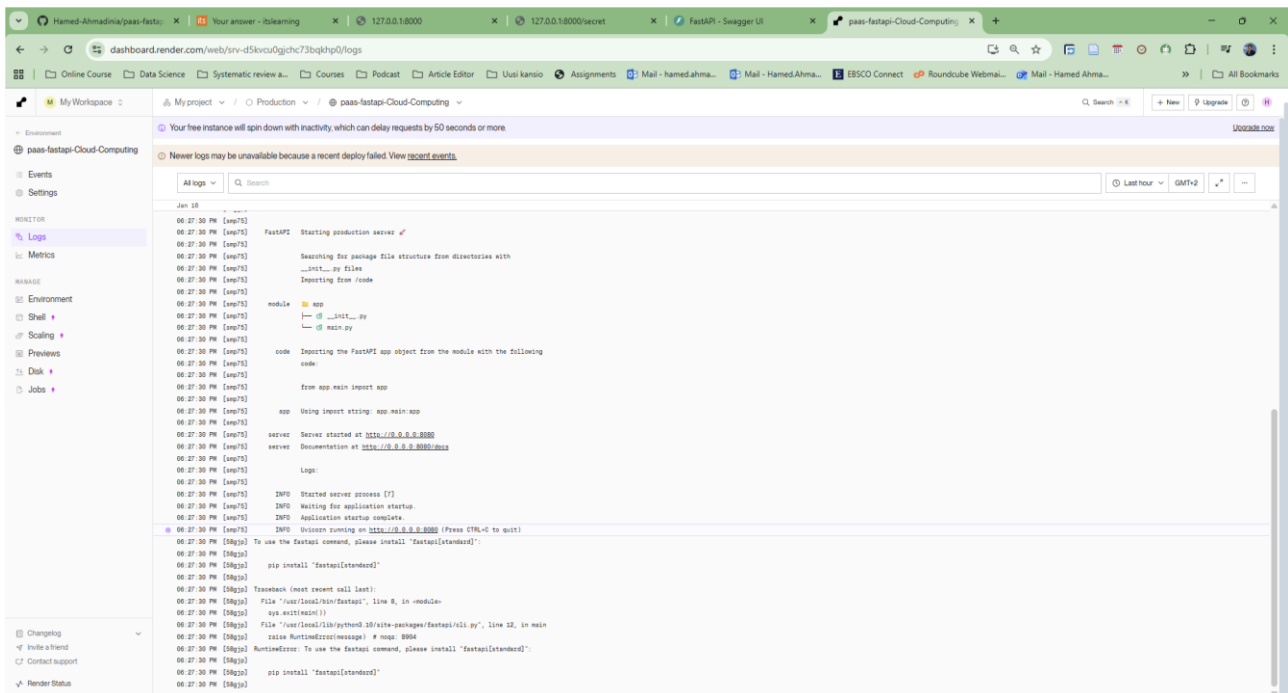
Resolution

These issues were resolved by updating the requirements.txt file to include fastapi[standard], modifying the Dockerfile to start the application using Uvicorn with the \$PORT environment variable, and redeploying the service after each configuration change. Ensuring that the API_KEY environment variable was correctly named and triggering a redeploy after saving the variable resolved the configuration error. After these changes, the application deployed successfully, and all endpoints behaved as expected.

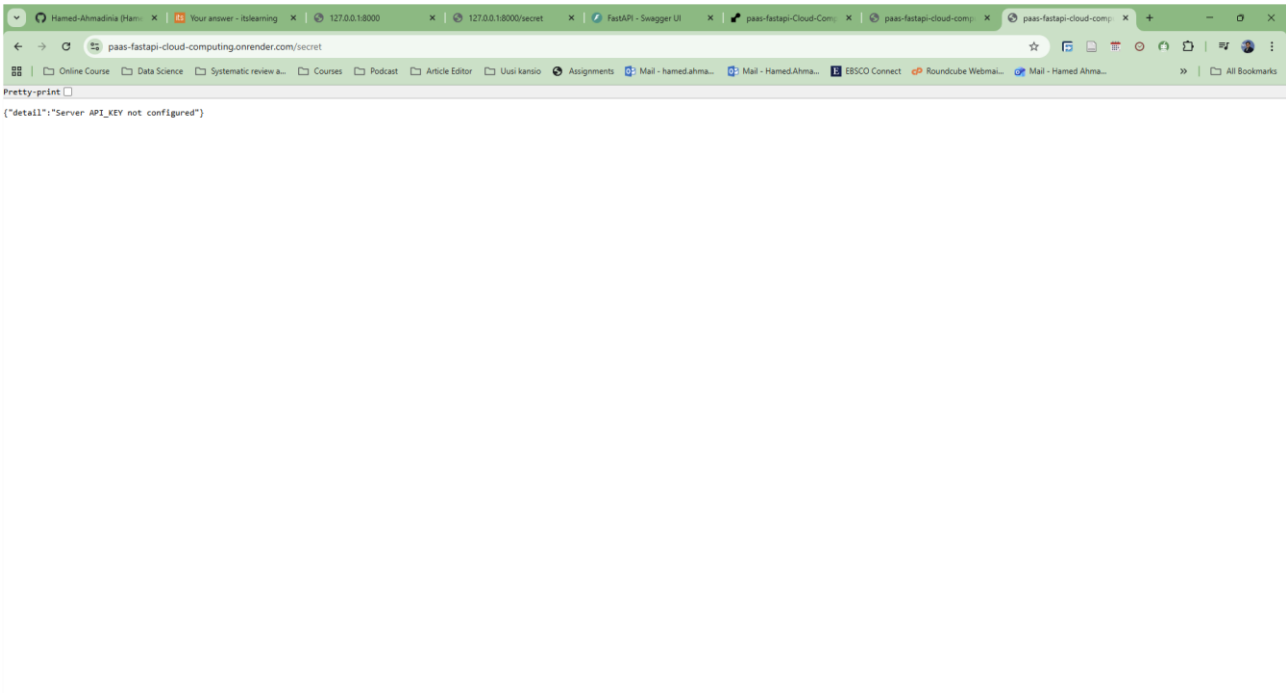
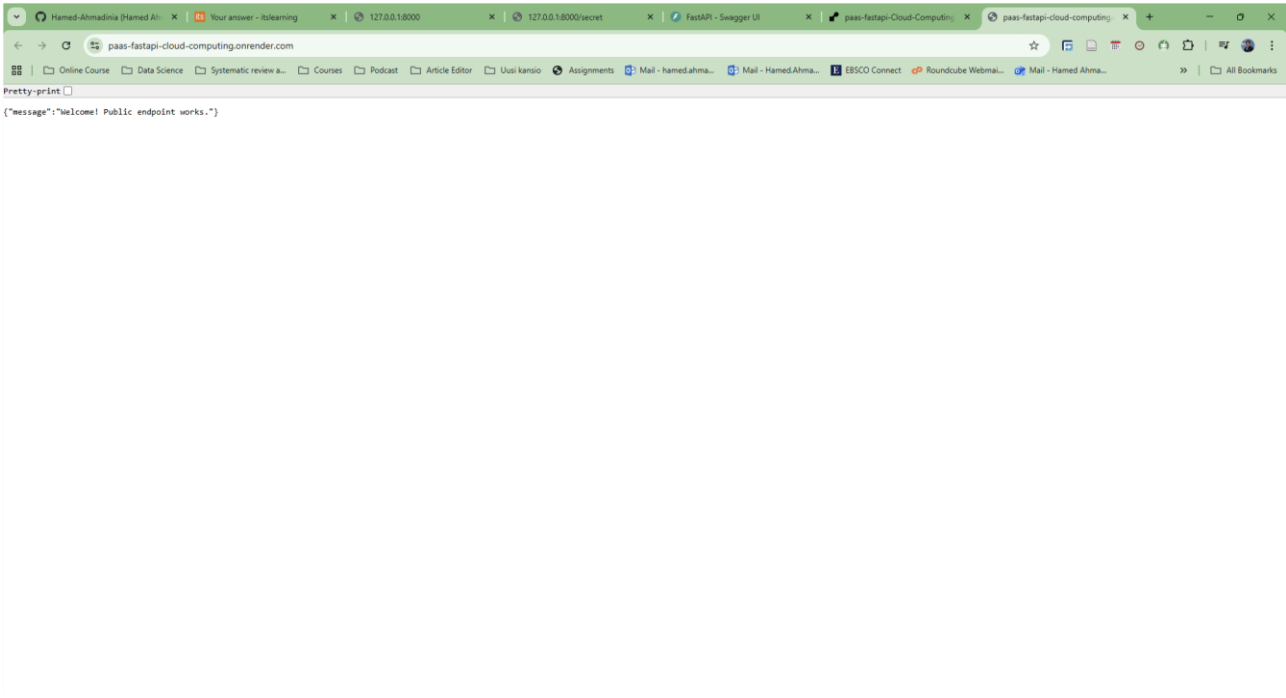
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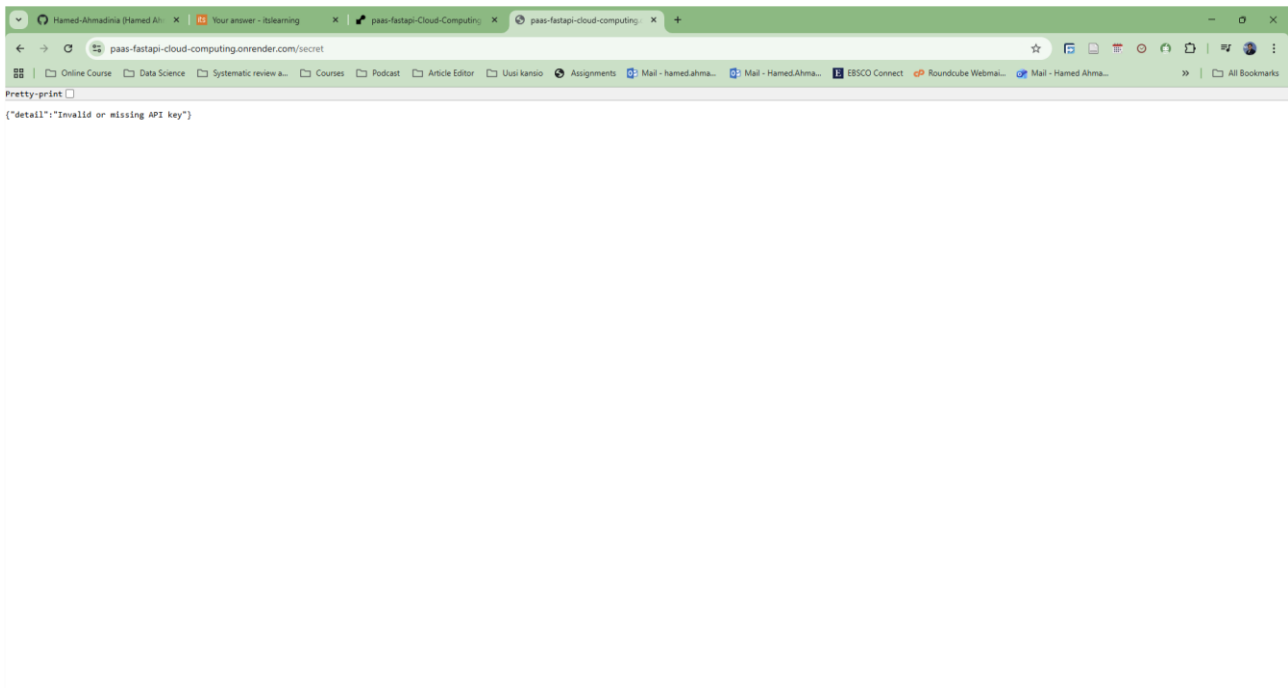
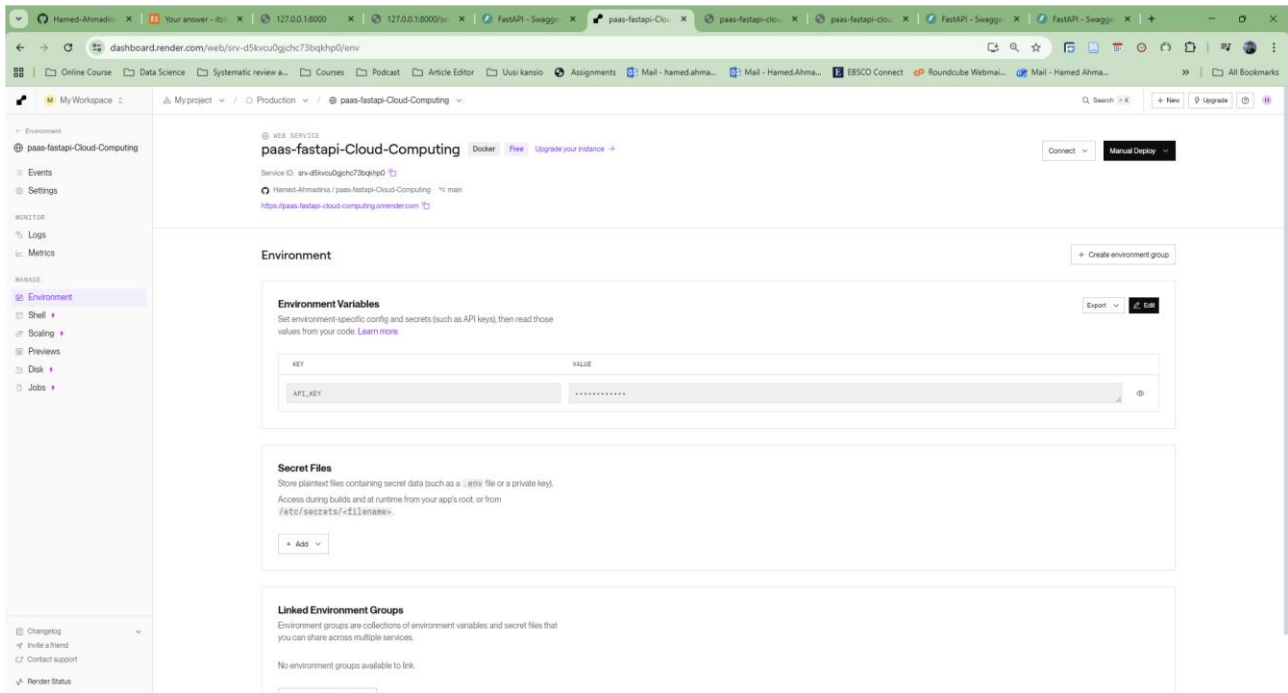
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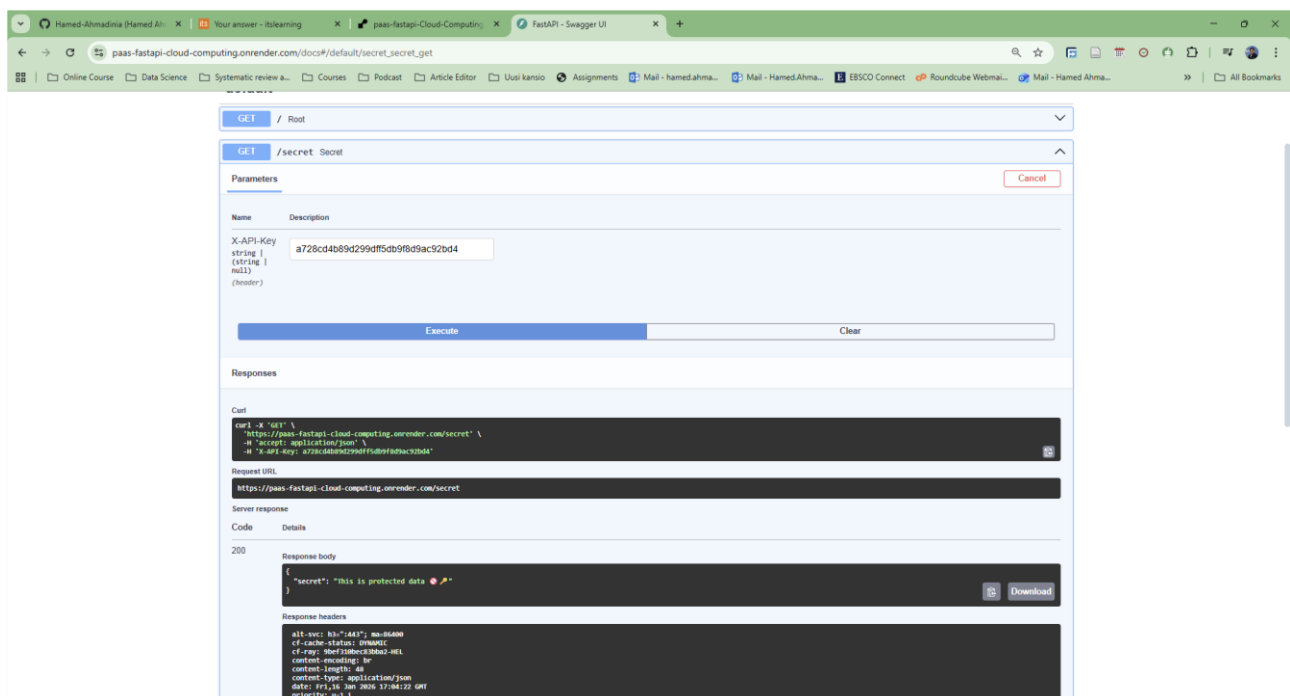
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Part C – JWT-Protected /messages Endpoint with Role-Based Authorization

Part C extends the API security model by implementing JSON Web Token (JWT) authentication with role-based authorization for the /messages endpoint. Unlike Parts A and B, which relied on API key validation, this section enforces identity-aware access control to ensure users can only access data permitted by their role, while administrators retain full access.

JWT support was added using the pyjwt library, which was included in requirements.txt. A fixed list of sample messages was defined directly in the application code, exactly as specified in the assignment instructions. Each message includes an id, user_id, and text, enabling authorization decisions based on ownership.

JWT verification was implemented using a FastAPI dependency named require_jwt. This dependency validates the presence and format of the Authorization header and enforces the Bearer token scheme. Invalid or missing tokens result in a 401 Unauthorized response. Valid tokens are decoded using the HS256 algorithm and a secret key loaded from the JWT_SECRET environment variable. The decoded token is then validated to ensure it contains the required sub and role claims.

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The /messages endpoint applies role-based authorization based on the decoded token. Users with the admin role receive the full list of messages, while users with the user role only receive messages where the user_id matches the sub claim. This ensures proper data isolation for regular users while allowing administrators unrestricted access.

JWT secrets were managed using environment variables. Locally, JWT_SECRET was stored in a .env file and loaded using python-dotenv, preventing sensitive data from being hardcoded or committed to version control. This approach is consistent with the secret management strategy used in Part B and aligns with standard production deployment practices.

Testing was performed using manually generated JWTs created via the jwt.io tool. Tokens were signed using the HS256 algorithm and the configured secret, with different sub and role values to represent regular users and administrators. The /messages endpoint was tested using PowerShell's Invoke-RestMethod, which provided reliable control over request headers.

Testing confirmed correct behavior in all scenarios: requests without a token returned 401 Unauthorized, user tokens returned only the user's own messages, and admin tokens returned all messages.

Issues Encountered and Resolution

Initial testing failed due to missing or incorrectly formatted Authorization headers and misconfigured environment variables, which prevented proper JWT decoding. Additional confusion arose when using the FastAPI Swagger UI, as header formatting inconsistencies caused repeated authorization failures.

These issues were resolved by enforcing the correct Authorization: Bearer <token> format, verifying proper loading of the JWT_SECRET variable, and switching to PowerShell-based testing. After these corrections, JWT verification and role-based access control worked as intended.

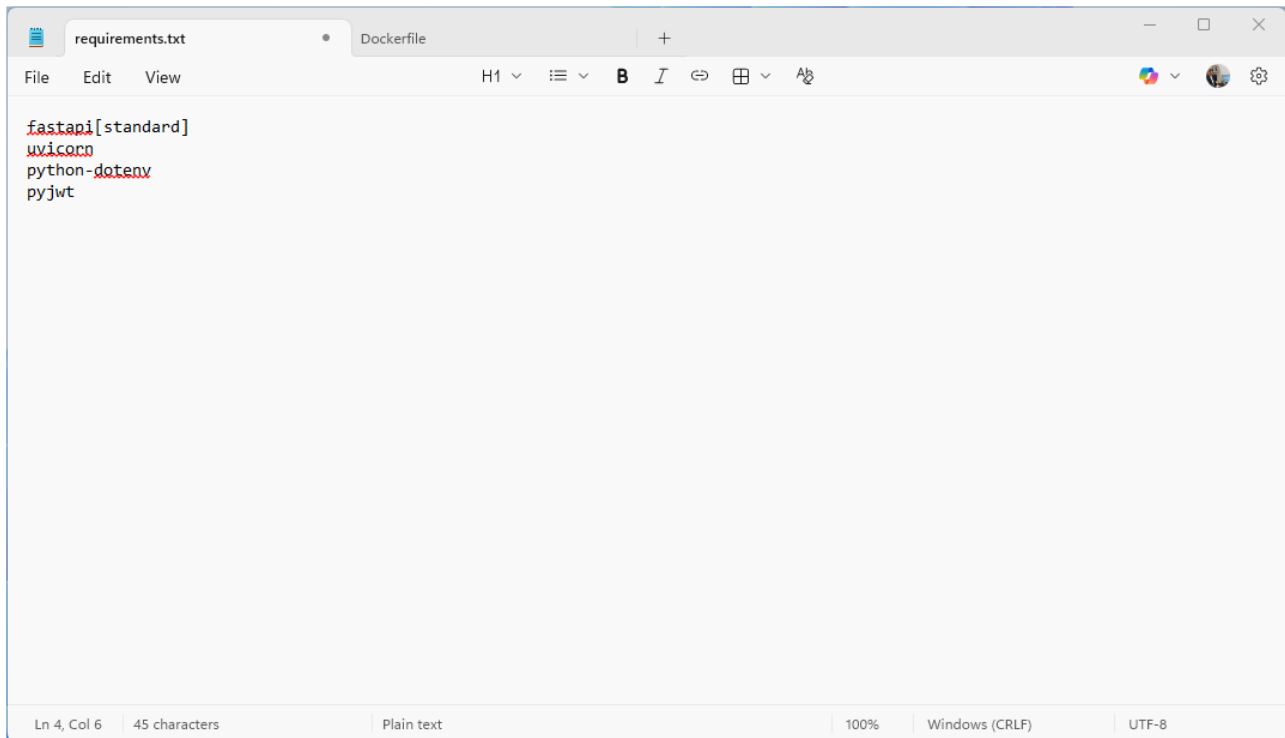
Conclusion

Part C successfully demonstrates JWT-based authentication and role-based authorization in a FastAPI application. By combining secure token validation, environment-based secret management, and claim-driven access control, the /messages endpoint enforces realistic and secure authorization rules that fully satisfy the optional assignment requirements.

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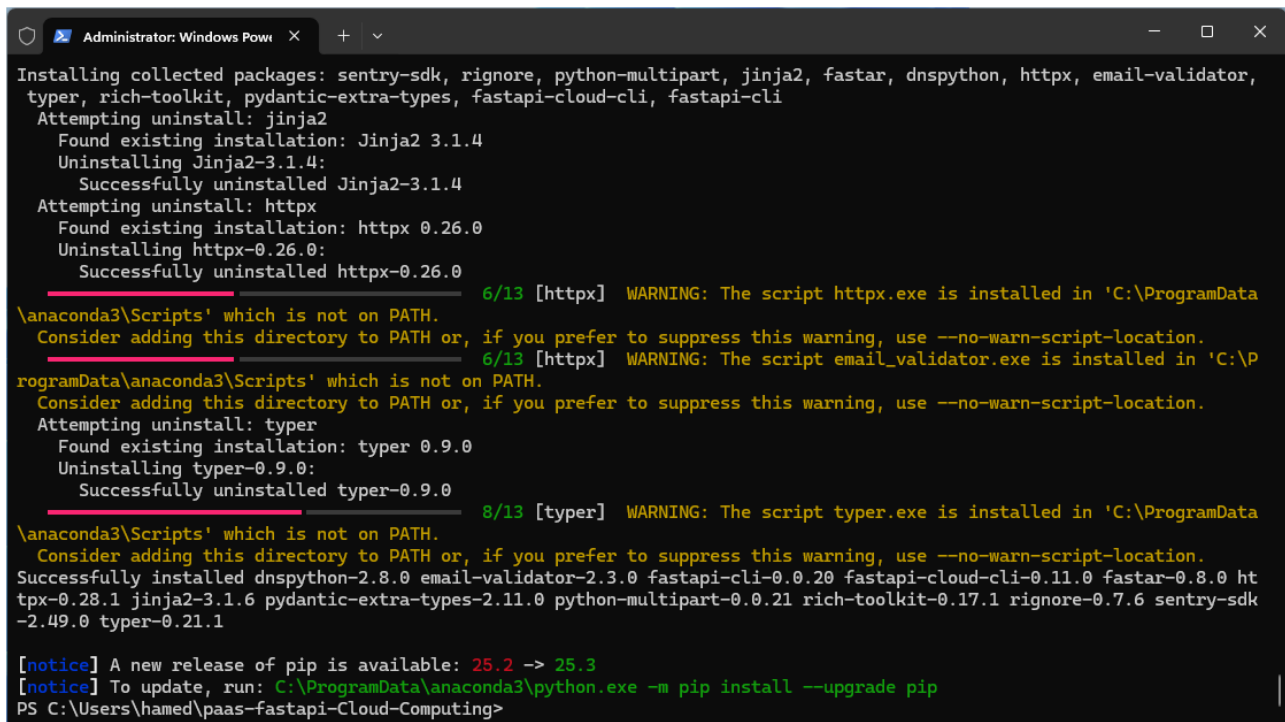
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The screenshot shows a code editor with two tabs: 'requirements.txt' and 'Dockerfile'. The 'requirements.txt' tab is active and contains the following text:

```
fastapi[standard]
uvicorn
python-dotenv
pyjwt
```

The editor interface includes a menu bar (File, Edit, View), a toolbar with various icons, and a status bar at the bottom indicating 'Ln 4, Col 6', '45 characters', 'Plain text', '100%', 'Windows (CRLF)', and 'UTF-8'.



The screenshot shows a Windows PowerShell terminal window titled 'Administrator: Windows PowerShell'. The output of a pip install command is displayed, showing the installation of various packages and warnings about script locations.

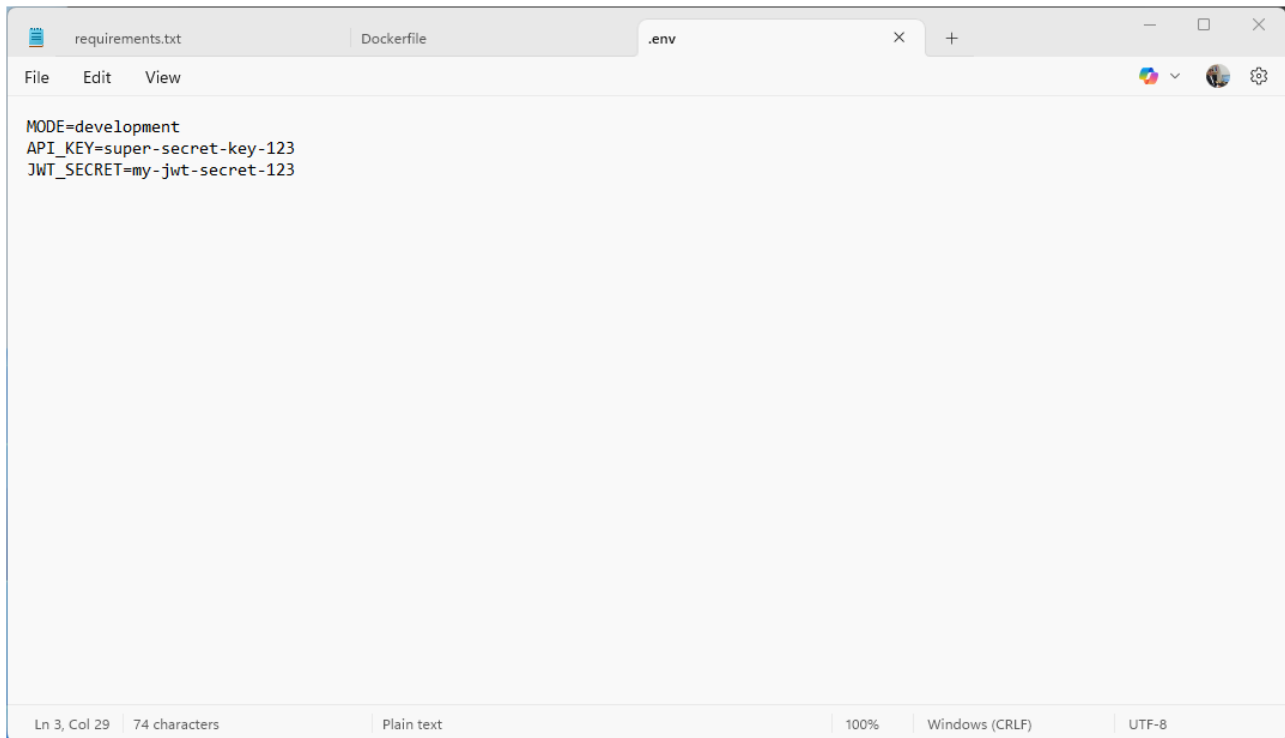
```
Installing collected packages: sentry-sdk, rignore, python-multipart, jinja2, fastar, dnspython, httpx, email-validator, typer, rich-toolkit, pydantic-extra-types, fastapi-cloud-cli, fastapi-cli
  Attempting uninstall: jinja2
    Found existing installation: Jinja2 3.1.4
    Uninstalling Jinja2-3.1.4:
      Successfully uninstalled Jinja2-3.1.4
  Attempting uninstall: httpx
    Found existing installation: httpx 0.26.0
    Uninstalling httpx-0.26.0:
      Successfully uninstalled httpx-0.26.0
  WARNING: The script httpx.exe is installed in 'C:\ProgramData\anaconda3\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
  WARNING: The script email_validator.exe is installed in 'C:\ProgramData\anaconda3\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
  Attempting uninstall: typer
    Found existing installation: typer 0.9.0
    Uninstalling typer-0.9.0:
      Successfully uninstalled typer-0.9.0
  WARNING: The script typer.exe is installed in 'C:\ProgramData\anaconda3\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed dnspython-2.8.0 email-validator-2.3.0 fastapi-cli-0.0.20 fastapi-cloud-cli-0.11.0 fastar-0.8.0 httpx-0.28.1 jinja2-3.1.6 pydantic-extra-types-2.11.0 python-multipart-0.0.21 rich-toolkit-0.17.1 rignore-0.7.6 sentry-sdk-2.49.0 typer-0.21.1

[notice] A new release of pip is available: 25.2 -> 25.3
[notice] To update, run: C:\ProgramData\anaconda3\python.exe -m pip install --upgrade pip
PS C:\Users\hamed\paas-fastapi-Cloud-Computing>
```

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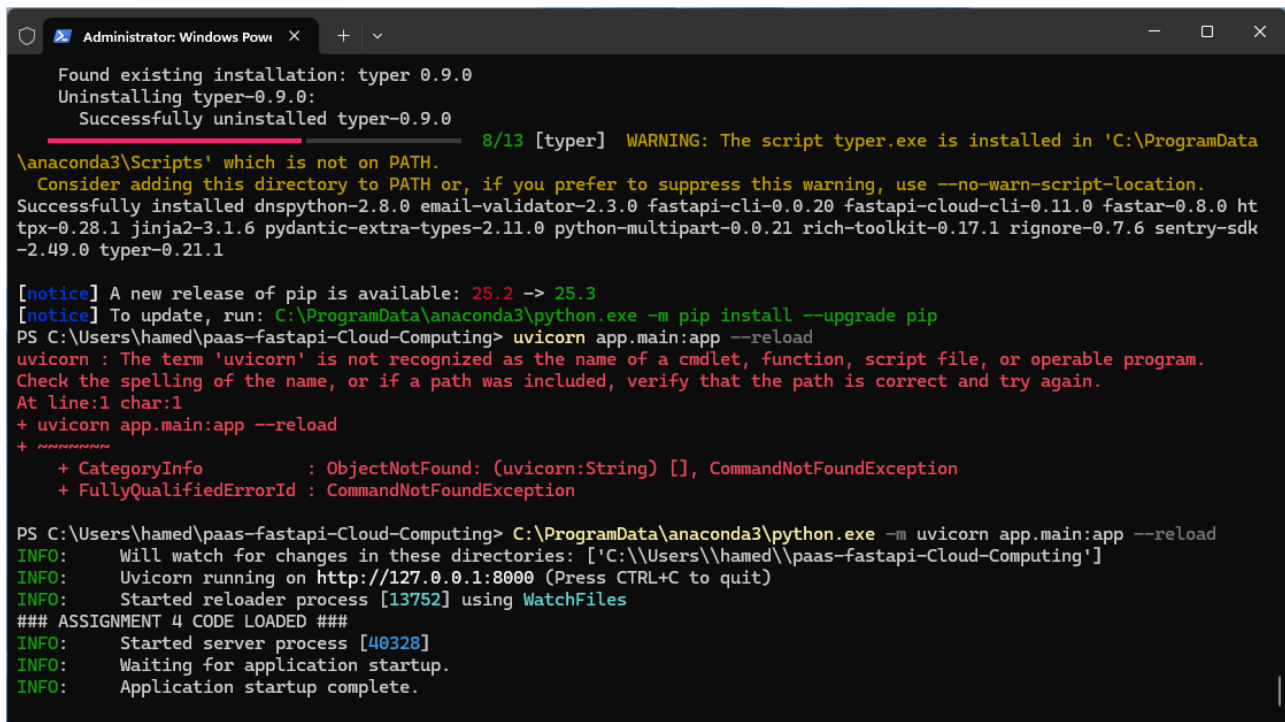
By: Hamed Ahmadinia



A screenshot of a code editor window with three tabs: 'requirements.txt', 'Dockerfile', and '.env'. The '.env' tab is active, showing the following content:

```
MODE=development
API_KEY=super-secret-key-123
JWT_SECRET=my-jwt-secret-123
```

The editor interface includes a menu bar with 'File', 'Edit', and 'View'. The status bar at the bottom indicates 'Ln 3, Col 29', '74 characters', 'Plain text', '100%', 'Windows (CRLF)', and 'UTF-8'.



A screenshot of a Windows PowerShell terminal window titled 'Administrator: Windows PowerShell'. The terminal shows the following commands and output:

```
Found existing installation: typer 0.9.0
Uninstalling typer-0.9.0:
  Successfully uninstalled typer-0.9.0
8/13 [typer] WARNING: The script typer.exe is installed in 'C:\ProgramData\anaconda3\Scripts' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this warning, use --no-warn-script-location.
Successfully installed dnspython-2.8.0 email-validator-2.3.0 fastapi-cli-0.0.20 fastapi-cloud-cli-0.11.0 fastar-0.8.0 ht
tpx-0.28.1 jinja2-3.1.6 pydantic-extra-types-2.11.0 python-multipart-0.0.21 rich-toolkit-0.17.1 rignore-0.7.6 sentry-sdk
-2.49.0 typer-0.21.1

[notice] A new release of pip is available: 25.2 -> 25.3
[notice] To update, run: C:\ProgramData\anaconda3\python.exe -m pip install --upgrade pip
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> uvicorn app.main:app --reload
uvicorn : The term 'uvicorn' is not recognized as the name of a cmdlet, function, script file, or operable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ uvicorn app.main:app --reload
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (uvicorn:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\Users\hamed\paas-fastapi-Cloud-Computing> C:\ProgramData\anaconda3\python.exe -m uvicorn app.main:app --reload
INFO: Will watch for changes in these directories: ['C:\\Users\\hamed\\paas-fastapi-Cloud-Computing']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [13752] using WatchFiles
### ASSIGNMENT 4 CODE LOADED ###
INFO: Started server process [40328]
INFO: Waiting for application startup.
INFO: Application startup complete.
```

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Course: Cloud Computing & Data Engineering

By: Hamed Ahmadiania

```
Administrator: Windows PowerShell

[{"id": 3, "user_id": 1, "text": "You have a new notification." },
{"id": 4, "user_id": 3, "text": "Password will expire in 5 days." },
{"id": 5, "user_id": 2, "text": "New login detected from a new device." },
{"id": 6, "user_id": 3, "text": "Your subscription has been updated." }
]

def require_jwt(authorization: str | None = Header(default=None, alias="Authorization")):
    if not authorization or not authorization.startswith("Bearer "):
        raise HTTPException(status_code=401, detail="Missing Bearer token")

    token = authorization.removeprefix("Bearer ").strip()
    secret = os.getenv("JWT_SECRET")
    if not secret:
        raise HTTPException(status_code=500, detail="Server JWT_SECRET not configured")

    try:
        payload = jwt.decode(token, secret, algorithms=["HS256"])
    except InvalidTokenError:
        raise HTTPException(status_code=401, detail="Invalid token")

    if "sub" not in payload or "role" not in payload:
        raise HTTPException(status_code=401, detail="Token missing required claims")

    return payload

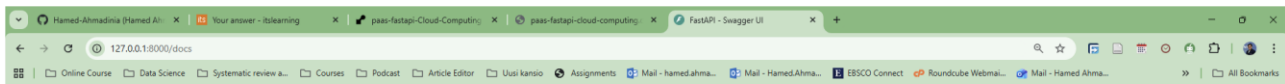
@app.get("/messages")
def messages(payload: dict = Depends(require_jwt)):
    role = payload["role"]
    sub = payload["sub"]

    if role == "admin":
        return MESSAGES

    try:
        sub_int = int(sub)
    except Exception:
        raise HTTPException(status_code=401, detail="Invalid sub claim")

    return [m for m in MESSAGES if m["user_id"] == sub_int]

PS C:\Users\hamed\paas-fastapi-Cloud-Computing>
```



FastAPI 0.104.1 0.104.1

default

- GET /
- GET /secret/secret
- GET /messages/messages

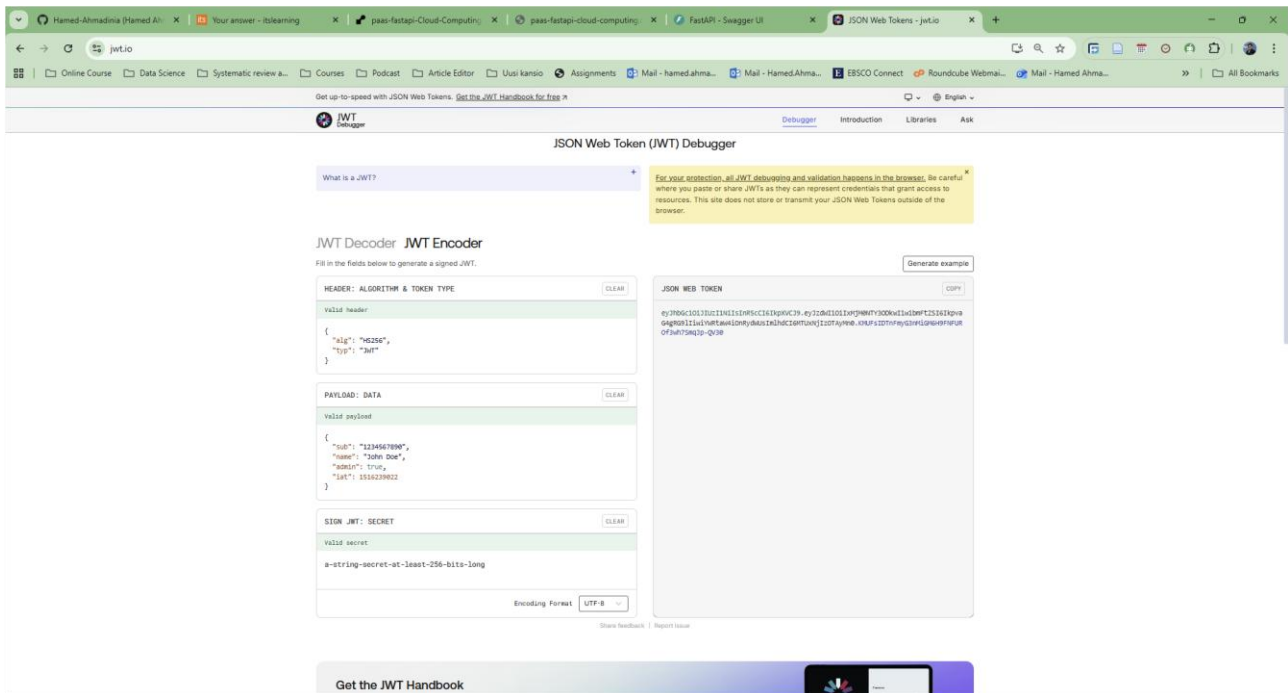
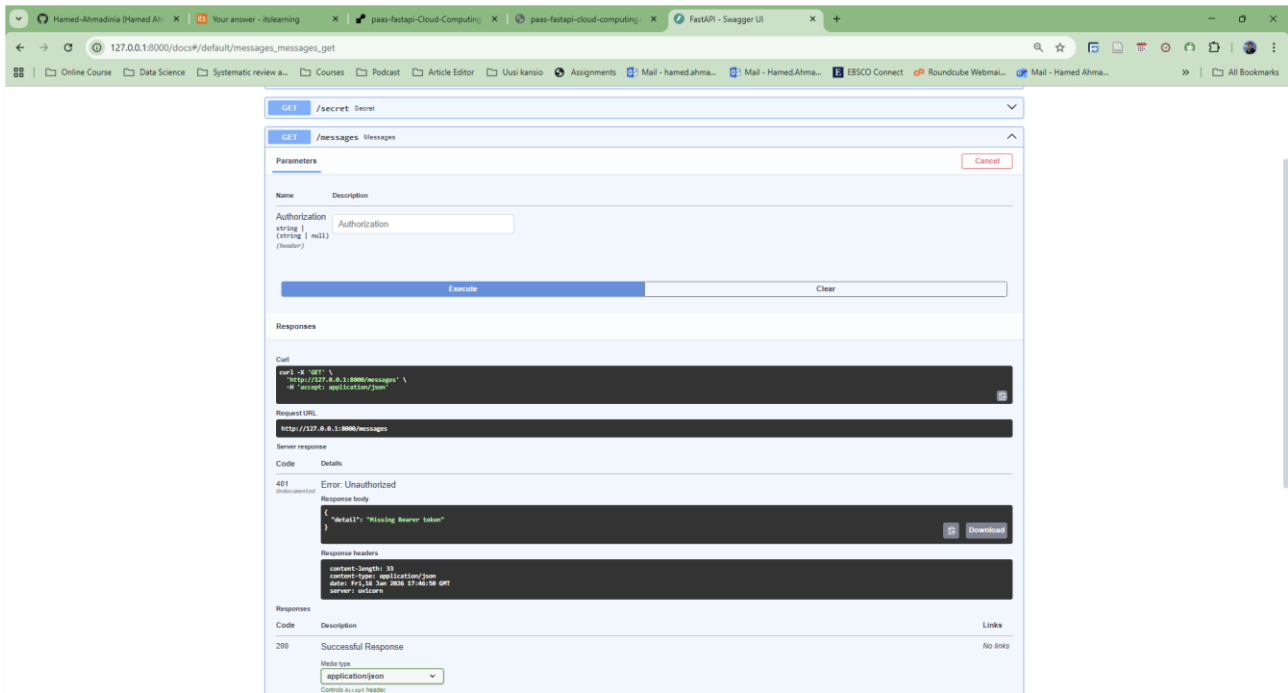
Schemas

- HTTPValidationError
- ValidationError

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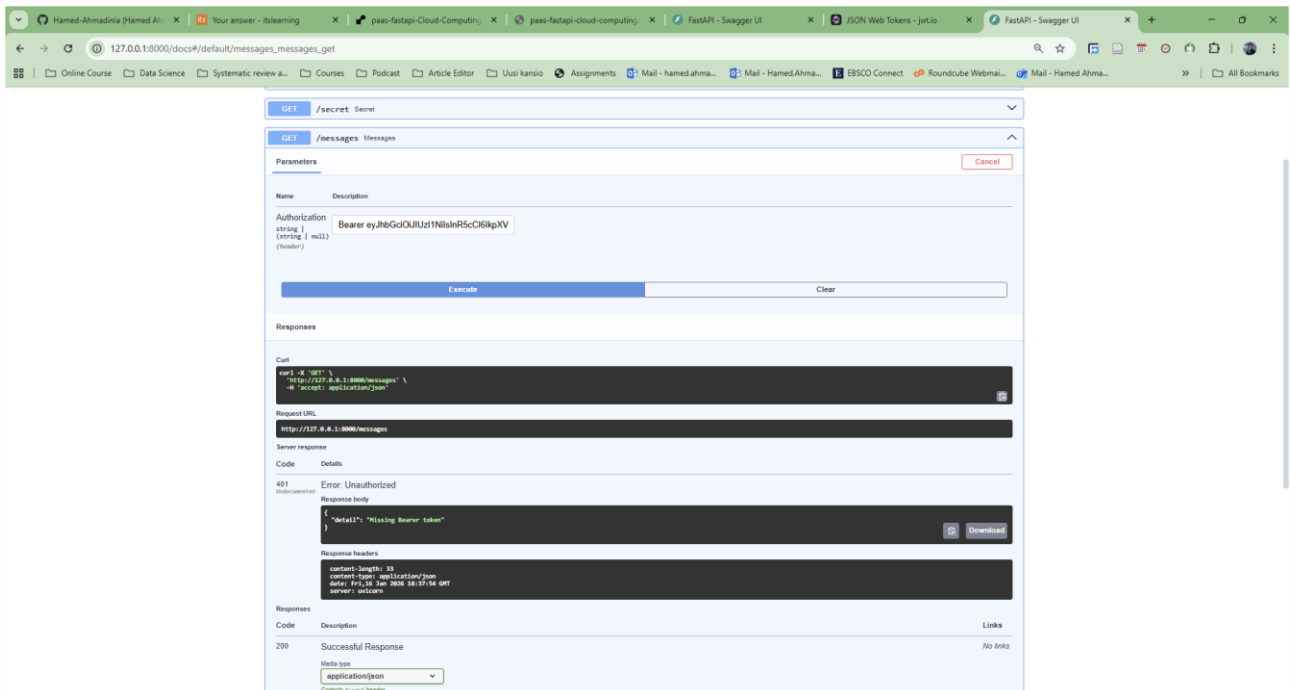
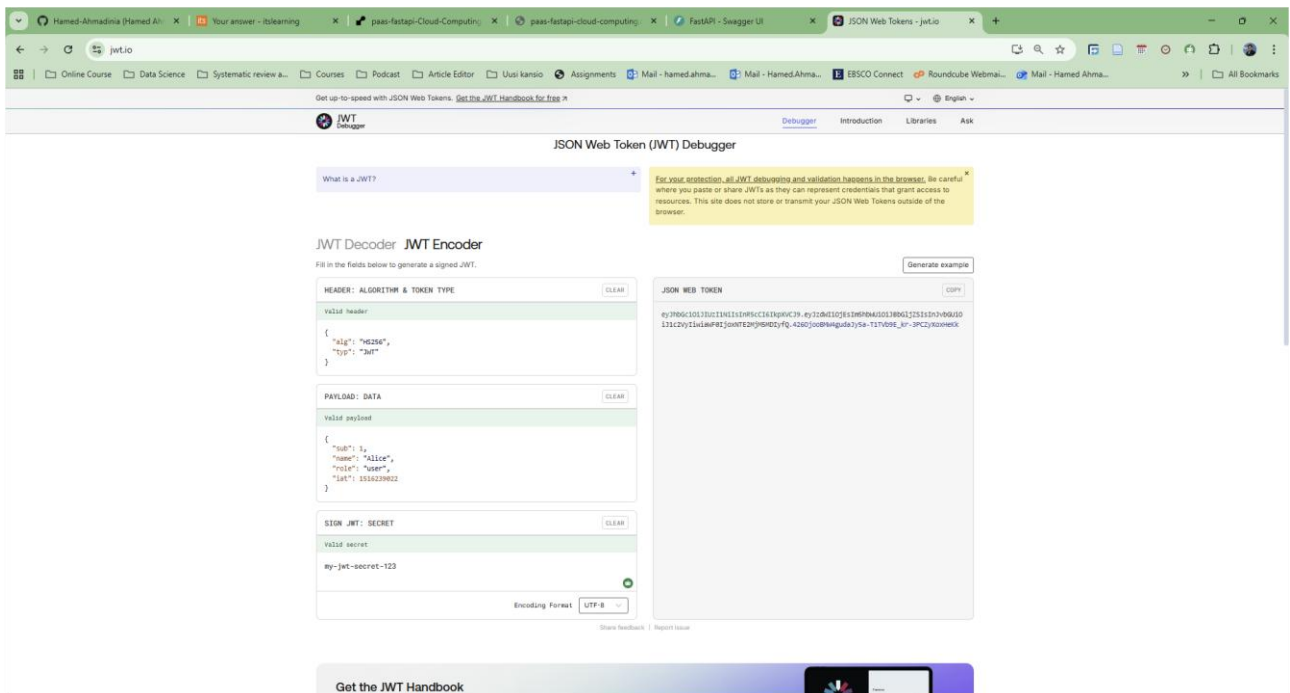
By: Hamed Ahmadiania



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By: Hamed Ahmadinia

```

Windows PowerShell
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> $token = "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJEsIm5hbWUiOiJBbGJzSIIsInJvbm90eSI6IjIc2VyIiwiaWF0IjoxNTE2MjM5MDIyYQ.4260jooBMW4gudaJy5a-T1TVb9E_kr-3PCZyXoxHeKk"
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> Invoke-RestMethod "http://127.0.0.1:8000/messages" -Headers @{ Authorization = "Bearer $token" }

id user_id text
--
1 1 Welcome to the platform!
3 1 You have a new notification.

PS C:\Users\hamed\paas-fastapi-Cloud-Computing> $adminToken = "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiJkSOSwibmFtPSI6IktFkbWluIiwicm9sZSI6ImFkbWluIiwiaWF0IjoxNTE2MjM5MDIyYQ.si6HTb3bgbEuReUzWU2n-0rnd0dYyJqp5SX5PsqeTpE"
PS C:\Users\hamed\paas-fastapi-Cloud-Computing> Invoke-RestMethod "http://127.0.0.1:8000/messages" -Headers @{ Authorization = "Bearer $adminToken" }

id user_id text
--
1 1 Welcome to the platform!
2 2 Your report is ready for download.
3 1 You have a new notification.
4 3 Password will expire in 5 days.
5 2 New login detected from a new device.
6 3 Your subscription has been updated.

PS C:\Users\hamed\paas-fastapi-Cloud-Computing>

```


Assignment 4: Assignment 4: Securing a Cloud API**Course:** Cloud Computing & Data Engineering**By:** Hamed Ahmadiania**Conclusion**

In this assignment, I successfully implemented and secured a FastAPI application using multiple authorization methods. By combining API key authentication and JWT-based authorization, I was able to control access to different endpoints based on both client authorization and user identity.

The assignment helped reinforce the importance of secure secrets management using environment variables in both local and cloud environments. Deploying the application to Render also highlighted the differences between development and production setups and the need for correct configuration when running cloud-based services.

Overall, this assignment provided practical experience in securing REST APIs and demonstrated how common security techniques are applied in real-world cloud applications.